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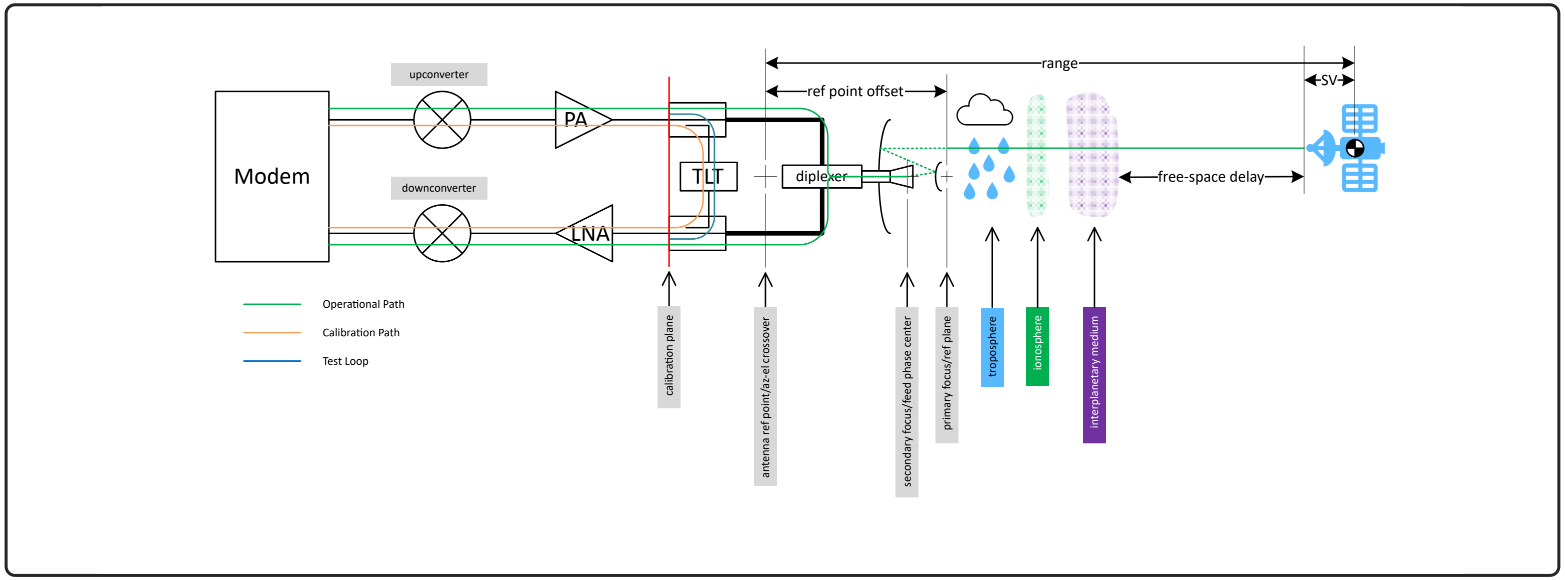
Delay Model

This page provides an overview of the processing required to convert the range delay measurements and other attributes provided by KSAT in the radiometric TDM file into actual range values that can be used for navigation and flight dynamics.

The range metric is a distance measurement between two physical points in space. On the ground side, the physical point is usually the intersection of the azimuth and elevation axis, as this point does not move relative to the aperture, regardless of the pointing angles of the antenna. The center of gravity of the vehicle is a reasonable choice on the SV side, since this point will be fixed within a navigation frame, regardless of the orientation of the vehicle. However, the choice of physical navigation point on the SV is up to the operator of the SV.

The “range” measurements performed by a KSAT antenna are really measurements of the round-trip signal delay from the modem output to the modem input. The delay due to range is a significant portion of this measurement, and is accounted for twice, since it is a round-trip measurement. As part of the process to convert the round-trip delay measurements to usable range values, multiple delay values that are not due to range need to be estimated and removed from the measurements.

This figure shows the signal path delays for the ground site, to the satellite, and back. Also shown are the calibration test path and the delays due to media, such as troposphere, ionosphere, and interplanetary medium. The ground site representation is notional, and individual ground antennas may vary from this model. However, in all cases, the delays in the ground antenna will be accounted for and included in the appropriate correction factor.



The calibration plane serves as the dividing line between those operational elements (like upconverters, downconverters, transmitter power amps, and low-noise amps) that are calibrated using the pre-pass calibration loop, and elements that are not part of this calibration and require external measurement or estimation. Furthermore, the calibration path includes items, specifically the test loop translator (TLT) that are not part of the operational path, and thus must be subtracted from the calibration value.

An accounting of the delays in this typical ranging scenario are listed in the table below. A fundamental concept of the way ranging data is collected by KSAT and presented to the user is that the TDM file contains raw measurements directly from the modem, along with correction factors to perform the required delay compensation. While this approach places more requirements on the user to perform post-processing of the data, it provides much more flexibility, and preserves the raw measurement values collected during the pass.

Delay values associated with the ground are provided in the TDM file. Other models, such as the spacecraft, are entirely the responsibility of the customer. Some models, such as troposphere, may be supported by information provided by the ground segment, though the actual choice and implementation of the model is the responsibility of the customer.

DELAY ELEMENT	OPS PATH	CAL PATH	FREQ DEP	PROVIDER
Transmit/Uplink				
modem transmitter	X	X	X	TDM (CORRECTION_RANGE)
IF/RF upconversion	X	X	X	TDM (CORRECTION_RANGE)
transmit amplifier	X	X	X	TDM (CORRECTION_RANGE)
tx test coupler-ops path	X			TDM (TRANSMIT_DELAY)
tx waveguide	X			TDM (TRANSMIT_DELAY)
diplexer	X			TDM (TRANSMIT_DELAY)
antenna feed	X			TDM (TRANSMIT_DELAY)
reflector	X			TDM (TRANSMIT_DELAY)
troposphere	X			Customer model
ionosphere	X		X	Customer model
interplanetary medium	X		X	Customer model
range delay	X			TDM (RANGE)
Turnaround Delay				

DELAY ELEMENT	OPS PATH	CAL PATH	FREQ DEP	PROVIDER
spacecraft	X			Customer model
Receive/Downlink				
range delay	X			TDM (RANGE)
interplanetary medium	X		X	Customer model
ionosphere	X		X	Customer model
troposphere	X			Customer model
reflector	X			TDM (RECEIVE_DELAY)
antenna feed	X			TDM (RECEIVE_DELAY)
diplexer	X			TDM (RECEIVE_DELAY)
rx waveguide	X			TDM (RECEIVE_DELAY)
rx test coupler-ops path	X			TDM (RECEIVE_DELAY)
receive LNA	X	X	X	TDM (CORRECTION_RANGE)
RF/IF downconversion	X	X	X	TDM (CORRECTION_RANGE)
modem receiver	X	X	X	TDM (CORRECTION_RANGE)
Test Loop				
tx test coupler-test path		X		site config file
test loop translator		X	X	site config file
rx test coupler-test path		X		site config file
Other				
reference point offset	X			site config file

Items shown as part of the operational (ops) path have to be estimated and subtracted from the delay measurement, except, of course, the actual range delay value, which is the whole point of the measurement. Items in the calibration (cal) path are measured during the pre-pass calibration process. Operational items that are also part of the calibration path are accounted for in the CORRECTION_RANGE entry in the TDM file. Items that are part of the calibration path, but not in the operational path (like the TLT), are subtracted from the calibration delay value prior to writing the CORRECTION_RANGE value to the TDM file.

Items shown as frequency dependent are likely to exhibit delays that are dependent on the carrier frequency. For customer-supplied models, it is up to the customer to compensate the chosen models for the appropriate frequency. For delays that are part of the calibration path, the transmitter and receiver should be configured for the appropriate frequencies prior to the calibration test. Finally, those that require manual estimation, such as the diplexer and TLT, are modelled using a delay vs. frequency table lookup and interpolation process. This table-driven interpolation model is implemented by the TDM generator software, and is accounted for when the TRANSMIT_DELAY or RECEIVE_DELAY values are written to the TDM file.